A Multi-UAV Cooperative Coverage Method Based on Sparse Dual-Attention Reinforcement Learning

Xiangwei Chen, Jian Yang, and Lidong Zhang

***Abstract*—This letter addresses the problem of cooperative multiple unmanned aerial vehicle (multi-UAV) coverage in dynamic environments, where UAVs aim to maximize target coverage while avoiding obstacles in real-time. Existing approaches often face difficulties in real-time decision-making for complex coverage tasks and exhibit limited adaptability. To overcome this problem, we propose a dual-attention mechanism with dynamic sparse activation, integrated into a multi-agent deep reinforcement learning (MADRL) framework under centralized training and decentralized execution (CTDE). Our method hierarchically processes the observation set—including targets, obstacles, and neighboring UAVs—while employing dynamic sparse activation to improve the accuracy and efficiency of key feature learning during training. This mechanism enables agents to selectively focus on critical environmental cues, reducing redundant computations and improving decision-making in real-time multi-UAV coordination. Furthermore, to enhance cooperative behavior among UAVs, an interaction attention mechanism is designed for each UAV, enabling them to coordinate coverage strategies through information exchange. Experimental results demonstrate that our approach outperforms existing methods in complex coverage tasks and exhibits strong scalability in large-scale UAV coordination scenarios.**

***Index Terms*—Dual-attention mechanism, multi-target coverage, deep reinforcement learning.**

I. Introduction

UNMANNED Aerial Vehicles (UAVs) possess several distinct advantages, including long endurance, high flexibility, compact size, and cost-effectiveness [1], [2]. Compared with single UAV systems, multi-UAV systems demonstrate enhanced fault tolerance and operational reliability, particularly in large-scale deployment scenarios [3]. These advantages significantly improve mission continuity and execution stability, rendering multi-UAV systems particularly suitable for executing complex tasks in dynamic environments, such as search and rescue operations, environmental monitoring, wildfire response, and urban surveillance [4], [5]. They also have potential in low-altitude urban counter-UAV (CUAV) missions to detect and neutralize malicious UAVs threatening other UAVs and ground assets [6].

In multi-UAV applications, UAVs must quickly reach key areas to handle urgent incidents [7], which can be formulated as real-time multi-target coverage tasks [8]. UAVs continuously monitor the environment, dynamically allocate targets, and navigate based on real-time data, aiming to minimize overlap, maximize coverage, and respond promptly to threats. Within limited observation ranges, UAVs must identify critical features, prioritize coverage tasks, and coordinate with neighbors to achieve effective multi-target coverage. Traditional centralized methods based on precomputed global plans often fail in dynamic scenarios. Distributed methods improve real-time responsiveness through local perception and autonomous decision-making, but their reliance on preset coordination rules and lack of data-driven learning can lead to resource conflicts and low efficiency in complex multi-target coverage tasks.

Recently, multi-agent deep reinforcement learning (MADRL) has emerged as an effective approach for complex multi-agent tasks. Under the centralized training and decentralized execution (CTDE) paradigm [9], [10], agents share observations during training but act based only on local observations during execution. In cooperative multi-UAV tasks, agents must consider both geometric relationships among entities and coordination strategies with other UAVs. Consequently, several studies have integrated MADRL with techniques such as attention mechanisms [11], [12], Transformers [13], Graph Neural Networks (GNNs) [14], Graph Attention Networks (GATs) [15], [16], Convolutional Neural Networks (CNNs) [17], [18]. These approaches effectively capture spatiotemporal relationships among entities and social interactions among agents, enhancing decision accuracy and adaptability in tasks like area coverage, target search, and tracking. Methods leveraging inter-agent communication [19], [20] and predictive modeling [21] further supplement local observations, offering valuable insights for our work.

However, in complex and dynamic environments, redundant observations can negatively affect the learning efficiency. It is crucial to extract the critical information from the observation. Several studies use attention mechanisms to evaluate temporal or spatial importance of information and inter-agent interactions [22], [23], [24]. However, they lack explicit redundancy

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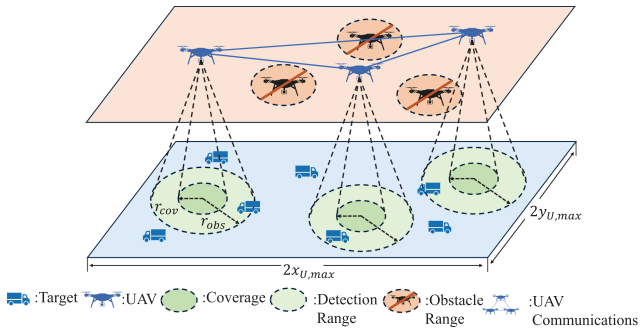

Fig. 1. Cooperative multi-UAV coverage scenario involving dynamic targets and obstacle regions.

suppression, leading to high-dimensional, computationally costly state representations that may impair decision-making.

Therefore, We propose a dual-attention method with dynamic sparse activation. Local observations of each agent are divided into targets, obstacles, and neighboring UAVs. An Entity Attention Module (EAM) extracts critical information from each set, while dynamic sparse activation progressively filters redundant data. An Interaction Attention Module (IAM) emphasizes key observations from neighboring UAVs to enhance cooperation. The proposed method is validated in multi-target coverage simulations of varying complexity.

The main contributions of this work are as follows:

- We propose a dual-attention learning method based on MADRL. In the EAM, entities are categorized into three sets, and a hierarchical attention mechanism is designed to extract critical information from each set.
- We introduce a dynamic sparse activation mechanism to effectively filter out redundant information. To avoid losing critical information and ensure proper filtering of redundant information, dynamic sparse regulation parameters are designed to improve the training efficiency.
- We construct a simulation environment to validate our method. Experimental results demonstrate that our method outperforms existing methods in scenarios with varying numbers of entities.

The rest of the letter is organized as follows. Section II formulates the system model and task objectives. Section III presents a dual-attention framework with sparsity based on MADRL. Section IV evaluates the proposed algorithm through simulation. Section V gives conclusions and future works.

II. System Model and Problem Formulation

A. Problem Statement

In this letter, we formulate the multi-target coverage problem in a two-dimensional environment consisting of N homogeneous UAVs $\mathbf{U} = \{1, \ldots, N\}$, M targets $\mathbf{G} = \{1, \ldots, M\}$, and K obstacle regions $\mathbf{T} = \{1, \ldots, K\}$, collectively referred to as entities $\mathbf{E} = \{\mathbf{U}, \mathbf{G}, \mathbf{T}\}$. As illustrated in Fig. 1, the positions of all entities are randomly initialized at the beginning of each mission. All UAVs share identical flight capabilities, with a fixed observation radius R_{obs} and coverage radius R_{cov}. The system objective is to maximize target coverage within a limited time horizon while ensuring collision avoidance.

B. System Model

We consider a dynamic environment consisting of N UAVs, M targets, and K obstacles. The UAVs are indexed as $i \in \{1, 2, \ldots, N\}$. The position of UAV U_i at time step t is denoted by $P_i^t = (x_i^t, y_i^t)$. The velocity and heading angle of U_i at time t are represented as v_i^t and θ_i^t, respectively. The kinematics of U_i is described as follows:

$$\begin{cases} x_i^t = x_i^{t-1} + v_i^{t-1} \cos\theta_i^{t-1} \Delta t & -x_{max} \le x_i^t \le x_{max} \\ y_i^t = y_i^{t-1} + v_i^{t-1} \sin\theta_i^{t-1} \Delta t & -y_{max} \le y_i^t \le y_{max} \\ v_i^t = v_i^{t-1} + a_i^{t-1} \Delta t & -v_{max} \le v_i^t \le v_{max} \end{cases} \tag{1}$$

Each UAV monitors entities within its sensor range. The observation indicator function for U_i is defined as follows:

$$\mathbb{I}_{\text{obs}}(U_i, E_j, t) = \begin{cases} 1 & \text{if } \|P_i^t - P_j^t\| \le R_{\text{obs}}, \\ 0 & \text{else.} \end{cases} \tag{2}$$

$\mathbb{I}_{obs}(U_i, E_j, t) = 1$ means that U_i has detected entity E_j. Similarly, the coverage indicator function is defined as:

$$\mathbb{I}_{cov}(U_i, G_m, t) = \begin{cases} 1 & \text{if } \|P_i^t - P_m^t\| \le R_{cov} \\ 0 & \text{else} \end{cases} \tag{3}$$

$\mathbb{I}_{cov}(U_i, G_m, t)$ describes that U_i fulfill the target coverage task when the distance between U_i and G_m is closer than R_{cov}. The objective is to maximize the cumulative target coverage over time:

$$J = \sum_{i=1}^{N} \sum_{t=0}^{T} \mathbb{I}_{cov}(U_i, G_m, t) \tag{4}$$

To avoid collisions, UAVs must maintain a safe distance from other UAVs and obstacles. Let R_{safe}^1 denote the minimum safe distance between UAVs, and R_{safe}^2 denote the minimum safe distance between UAVs and the boundaries of obstacles. UAVs must satisfy the following constraints:

$$\begin{cases} \|P_i^t - P_j^t\| \ge R_{safe}^1 & \forall U_i, U_j \in U, i \ne j \\ \|P_i^t - P_k^t\| \ge R_{safe}^2 & \forall U_i \in U, \forall T_k \in T \end{cases} \tag{5}$$

C. Problem Transformation

The cooperative multi-UAV target coverage task is formulated as a decentralized partially observable Markov decision process (Dec-POMDP), defined by the tuple $\langle N, S, O, A, P, R, \gamma \rangle$, where N denotes the number of UAVs. The global state is $s \in S = \mathbb{R}^{N \times D}$, with D representing the state dimensionality. Each UAV U_i receives a local observation $o_i = O(s) \in \mathbb{R}^d$, where $d < D$. The action of U_i is $a_i \in A$, and the joint action of all UAVs is denoted by $\mathbf{a} = (a_1, \ldots, a_N)$. The state transition dynamics are governed by $P(s' \mid s, \mathbf{a})$. All UAVs share a common reward function $r : S \times A \to \mathbb{R}$, and γ is the discount factor. Each UAV follows a decentralized policy $\pi_i^\theta(a_i \mid o_i)$ and aims to maximize the expected discounted return $G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k}$.

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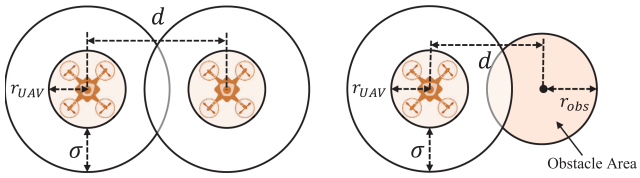

Fig. 2. Conflict threat zones between UAVs and between UAVs and obstacles.

Observation Space: The observation o_i^t for U_i at time t consists of four parts: 1) The state of U_i itself, denoted as $[P_i^t, v_i^t]$, where $P_i^t = (x_i^t, y_i^t)$, and $v_i^t = (v_i^{x,t}, v_i^{y,t})$ denote the position and velocity of U_i, respectively. 2) The relative state information of other UAVs is denoted as $[P_j^t - P_i^t, v_j^t - v_i^t]$, for $j \leq N$ and $j \neq i$. 3) The relative state information of targets G_m is given as $[P_m^t - P_i^t]$, where $m \leq M$. 4) The relative state information of all obstacles T_k, denoted as $[P_k^t - P_i^t, r_k]$, where $k \leq K$ and r_k denotes the radius of obstacle T_k. For the entities outside the observation range of U_i, the relative information would be set to zero vectors.

Action Space: The action of U_i consists of the acceleration in the x direction and y direction, denoted as $a_i^t = (a_i^{t,x}, a_i^{t,y})$, where $a_i^{t,x}$ and $a_i^{t,y} \in [-a_{max}, a_{max}]$. The policy $\pi_i^\theta(a_i|o_i)$ outputs normalized actions in $[-1, 1]$, which are scaled by a_{max} to obtain actual values.

Reward Function: The reward function is decomposed into two key components: collision penalty and coverage reward.

1) Collision Penalty: As shown in Fig. 2, we define a threat zone of width σ around each UAV. If the distance d between two UAVs is less than $2r_{UAV} + \sigma$, a potential collision risk exists. We define the collision risk penalty function as follows:

$$r_1 = \begin{cases} -\frac{0.02}{e^{d-2r_{UAV}}} & 2r_{UAV} < d < 2r_{UAV} + \sigma \\ -50 & 0 \leq d \leq 2r_{UAV} \\ 0 & \text{else} \end{cases} \tag{6}$$

Similarly, the penalty function for UAVs approaching obstacles is defined as follows:

$$r_2 = \begin{cases} -\frac{0.02}{e^{d-D}} & D < d < D + \sigma \\ -50 & 0 \leq d \leq D \\ 0 & \text{else} \end{cases} \tag{7}$$

where $D = r_{UAV} + r_{obs}$.

2) Coverage Reward: Each UAV is awarded 10 reward units for successfully covering a target, denoted as $r_3 = 10$.

In summary, the overall reward function for each UAV can be expressed as:

$$R_{UAV} = r_1 + r_2 + r_3 \tag{8}$$

III. The Proposed Algorithm

A. Overall Framework

The framework of our method is illustrated in Fig. 3. The approach is based on the CTDE paradigm.

During centralized training, the critic network incorporates an Interaction Attention Module (IAM) based on multi-head attention (MHA) to selectively attend to observations from other agents, thereby extracting interaction information most relevant to the target agent. In both the actor and critic networks, an Entity Attention Module (EAM) categorizes entities into three types and applies a three-layer self-attention structure to capture salient features within each category. In addition, a dynamic sparse activation mechanism is introduced to sparsify attention weights, effectively filtering redundant information. During decentralized execution, the critic network is removed, and each agent relies solely on its actor network, using local observations to generate actions for autonomous decision-making.

B. Entity Attention Module (EAM)

For each agent i, the observation-action pair (o_i, a_i) is decomposed into four components according to different entity sets: 1) Own state-action pair (o_i^{own}, a_i), which includes the position, velocity, and action information of agent i; 2) Target state information o_i^{goal}, which contains the relative position of the target; 3) Teammate state information o_i^{ally}, which consists of the relative positions and velocities of the other agents; and 4) Obstacle state information $o_i^{obstacle}$, which includes the relative positions and radius of the obstacles. In the EAM, (o_i^{own}, a_i) is used as the query, while o_i^{goal}, o_i^{ally}, and o_i^{obstacle} serve as keys and values, with their relevance measured by the scaled dot-product attention.

$$\text{score}_i^{\text{entity}_j} = \frac{\left(o_i^{\text{entity}_j}\right)^T W_{\text{entity},k} W_{\text{own},q}(o_i^{\text{own}}, a_i)}{\sqrt{d_{\text{EAM}}}} \tag{9}$$

where entity $\in$ [ally, goal, obstacle] and d_{EAM} denotes the dimensionality of $W_{\text{entity},k}$. Subsequently, the computed similarities are normalized into a probability distribution using an activation function. Although Softmax is commonly adopted for this purpose, it is inherently non-sparse and assigns non-zero attention weights to all entities, regardless of their relevance. This property is suboptimal in scenarios where many observed entities are of negligible importance and may lead to information redundancy. To address this issue, we introduce a sparse activation function that enables the agent to focus on the most relevant entities by assigning zero attention weights to less important ones.

The sparse activation function [25] is defined as follows:

$$\text{sparsemax}(z) := \arg\min_{\boldsymbol{p} \in \Delta^d} \|\boldsymbol{p} - \boldsymbol{z}\|^2 \tag{10}$$

where Δ^d denotes the d-dimensional probability simplex, defined as $\Delta^d := \{p \in \mathbb{R}^d : p \geq 0, \|p\|_1 = 1\}$. By sorting the input vector $\boldsymbol{z}$ in descending order and applying the Lagrangian method, the solution to the Sparsemax function is derived as follows:

$$\text{Sparsemax}_i(\boldsymbol{z}) = [z_i - \tau(\boldsymbol{z})]_+ \tag{11}$$

$$\text{s.t.} \quad \sum_j [z_j - \tau(\boldsymbol{z})]_+ = 1 \tag{12}$$

$$\tau(\boldsymbol{z}) = \frac{\left(\sum_{j \leq k(\boldsymbol{z})} z_{(j)} - 1\right)}{k(\boldsymbol{z})} \tag{13}$$

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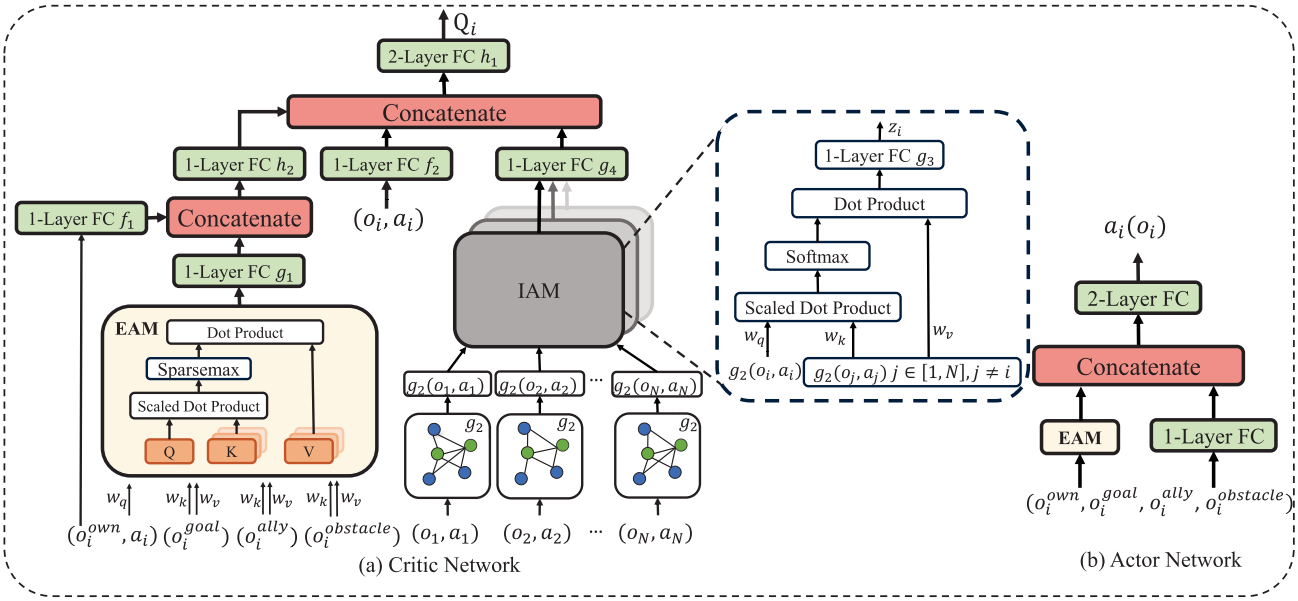

Fig. 3. The overall framework of the Sparse Dual-Attention Multi-Agent Twin Delayed Deep Deterministic Policy Gradient (SDA-MATD3). (a) The critic network of agent i, where EAM denotes the Entity Attention Module and IAM denotes the Interaction Attention Module. (b) The actor network of agent i.

$$k(\boldsymbol{z}) := \max\left\{k \in [K] \mid 1 + kz_{(K)} > \sum_{j \le k} z_{(j)}\right\} \tag{14}$$

where $k(\boldsymbol{z})$ denotes the number of non-zero elements in the solution, and $[\cdot]_+ = \max\{0, \cdot\}$. $\tau(\boldsymbol{z})$ denotes the threshold function for $\boldsymbol{z}$, such that if the probability z_j is less than $\tau(\boldsymbol{z})$, z_j is set to zero.

However, applying the Sparsemax function from the early training stage may incorrectly suppress critical entities due to its strong sparsity, since the agents have not yet learned the relative importance of different observations, which can hinder training. To mitigate this issue, we introduce a dynamic sparsity adjustment mechanism that gradually regulates the sparsity level during training. Specifically, the Softmax function is employed in the early stage to enable the EAM to learn from a broader range of entity information. As training progresses, the Sparsemax function is progressively incorporated to enforce sparsity, ensuring that only non-critical entities are filtered after the EAM has learned to focus on key information. A sparsity control parameter λ is introduced to govern this transition and is increased linearly over the training process. The attention weight of agent i on entity j within its observation range is defined as:

$$\begin{aligned} w_i^{\text{entity}_j} &= \lambda \cdot \text{sparsemax}\left(\text{score}_i^{\text{entity}_j}\right) \\ &\quad + (1-\lambda) \cdot \text{softmax}\left(\text{score}_i^{\text{entity}_j}\right) \end{aligned} \tag{15}$$

Subsequently, for each entity set, the attention weight $w_i^{\text{entity}_j}$ is multiplied by the corresponding value vector $v_i^{\text{entity}_j}$ to obtain the attention value z_i^{entity}. These attention values from different entity sets are then concatenated. The final output of the EAM is obtained through a fully connected layer, denoted as g_1.

$$\text{att}_i^{\text{EAM}} = g_1\left(\text{concat}(z_i^{\text{entity}})\right) \tag{16}$$

$$z_i^{\text{entity}} = \sum_{j \in [1,M]} w_i^{\text{entity}_j} v_i^{\text{entity}_j} \tag{17}$$

Similarly, the EAM is applied to the actor network. Since the actor does not take action information as input, o_i^{own} is used as the query, while the remaining implementation follows the same procedure as in the critic network.

C. Interaction Attention Module (IAM)

During the centralized training, for each agent i, the IAM takes $(o_j, a_j), j \in [1, N]$ as input into an encoder g_2, which consists of fully connected layers, to generate the embedding $g_2(o_j, a_j)$ for each observation-action pair. The IAM utilizes the parameter matrix W_q to generate the query based on $g_2(o_i, a_i)$, and employs W_k and W_v to generate the key and value from $g_2(o_j, a_j), j \neq i$. The similarity between the embedding of agent i and other agents is computed using the scaled dot product and normalized through the Softmax function:

$$\text{score}_{i,j} = \frac{g_2(o_j, a_j)^T W_k^T W_q g_2(o_i, a_i)}{\sqrt{d_{IAM}}} \tag{18}$$

$$w_{i,j} = \frac{\exp(\text{score}_{i,j})}{\sum_{k \in [1,N], k \neq i} \exp(\text{score}_{i,k})} \tag{19}$$

The attention weights for agent i, $w_{i,j}$, $j \in [1, N]$, $j \neq i$, are multiplied by the corresponding value vectors (excluding agent i) v_j, $j \in [1, N]$, $j \neq i$, to obtain the attention values for agent i. These attention values are then passed through a fully connected

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layer g_3 to obtain z_i:

$$z_i = g_3\left(\sum_{j\in[1,N],j\neq i} w_{i,j}v_j\right) \tag{20}$$

where $v_j = W_v g_2(o_j, a_j)$. The outputs of the IAM, $[z_1,\ldots,z_N]$, are concatenated and passed through another fully connected layer g_4 to obtain the final output:

$$\text{att}_i = g_4\left(\text{concat}(z_k, k\in[1,N])\right) \tag{21}$$

Finally, under the dual attention mechanism, the centralized Q-function for agent i is expressed as:

$$Q_i = h_1\left(h_2\left(f_1(o_i^{\text{own}}, a_i), \text{att}_i^{EAM}\right), f_2(o_i, a_i), \text{att}_i\right) \tag{22}$$

D. Training Algorithm

Our method is implemented based on the multi-agent twin delayed deep deterministic policy gradient (MATD3) algorithm, an optimized version of the multi-agent deep deterministic policy gradient (MADDPG) [26] algorithm.

The critic network estimates the Q-values by aggregating the observation-action information from all UAVs. The loss function for training the critic network, parameterized by $\theta_{i,j}$, is defined based on the temporal difference (TD) error:

$$\mathcal{L}(\theta_{i,j}) = \frac{1}{B}\sum_{k=1}^{B}\left[\left(Q_{i,j}(\mathbf{o}^k, \mathbf{a}^k; \theta_{i,j}) - y_{i,j}^k\right)^2\right] \tag{23}$$

$$y_{i,j} = r_i + \gamma \min_{j=1,2} Q'_{i,j}\left(\mathbf{o}', \mathbf{a}'; \theta'_{i,j}\right)\Big|_{\mathbf{a}'=\boldsymbol{\mu}'_{\boldsymbol{\varphi}'}(\mathbf{o}')+\boldsymbol{\epsilon}} \tag{24}$$

The actor network selects the optimal actions by maximizing the Q-values estimated by the critic network. The gradient of the actor network is computed as follows:

$$\nabla_{\varphi_i} J \approx \frac{1}{B}\sum_{k=1}^{B} \nabla_{\varphi}\mu_{\varphi_i}(o_i^k)\cdot \nabla_{a_i^k} Q_{i,1}(\mathbf{o}^k, \mathbf{a}^k; \theta_{i,j})\big|_{a_i^k=\mu_{\varphi_i}(o_i^k)} \tag{25}$$

where $\theta_{i,j}$ and $\theta'_{i,j}$ denote the parameters of the j-th ($j = 1, 2$) critic network and the target critic network for U_i, respectively, φ_i and φ'_i denote the parameters of the actor network and target actor network for U_i, respectively, $Q'_{i,j}$ denotes the Q-value of the j-th target critic network for U_i, $\mathbf{o}'$ denotes the local observations of all UAVs at the next state, and $\mathbf{a}'$ denotes the actions generated by the target actor network for the next state, with added noise.

The target network is updated via soft updates, as follows:

$$\theta'_{i,j} \leftarrow \tau\theta_{i,j} + (1-\tau)\theta'_{i,j} \tag{26}$$

$$\varphi'_i \leftarrow \tau\varphi_i + (1-\tau)\varphi'_i \tag{27}$$

where $\tau \ll 1$.

The training algorithm is detailed in Algorithm 1.

Algorithm 1: Training Algorithm for SDA-MATD3.

```
Initialize memory D and network parameters θ_{i,j}, θ'_{i,j},
  φ_i, φ'_i, the actor network delay frequency f_a;
for epoch ← 1, 2, ··· do
    Reset the environment and receive the initial state
      observations o = (o_1, ..., o_N);
    for timestep ← 1, 2, ··· do
        Select and execute action a_i = μ_{φ_i}(o_i) based
          on the current actor network for each UAV;
        Receive rewards r = (r_1, ..., r_N) and next
          observations for each UAV o' = (o'_1, ..., o'_N);
        Store the transition ⟨o, a, r, o'⟩ into memory D;
        Randomly sample a minibatch from D;
        for UAV i ← 1 to N do
            Update the critic network according to
              Equation (23) and (24);
            if f_d then
                Update the actor network according to
                  Equation (25);
                Update the target actor network
                  according to Equation (27);
            Update the target critic network according
              to Equation (26);
```

IV. Experiments

A. Experimental Setup

The environment is confined within a square area with boundaries ranging in $[-1, 1]$. Each UAV has identical configurations: a body radius of 0.05 units, a detection radius of 0.6 units, and a coverage radius of 0.15 units. The simulation operates with a time step of 0.03 seconds, and the maximum UAV speed is 0.05 units per time step. The targets and obstacles move in random directions with random speeds. The maximum speed for the targets is set to 0.02 units per time step, and for the obstacles, it is 0.01 units per time step. To maintain a consistent level of environmental complexity, once a target is covered by a UAV, it is removed from the environment and a new target is randomly generated at a different location.

B. Simulation Results Analysis

To evaluate the performance of our method (SDA-MATD3), we compare it with three baselines: the standard MATD3, the multi-actor-attention-critic based MATD3 (MAAC-MATD3) [22], and the global-and-local attention based MATD3 (GLA-MATD3) [23]. Fig. 4 illustrates the network structures of these two attention-based baselines. Each method is trained for 100,000 episodes, with each episode consisting of 500 time steps.

Fig. 5 illustrates the learning curves under three different environmental settings. Compared with the MATD3 baseline, incorporating the attention module significantly improves UAV coverage performance. Our method consistently outperforms the

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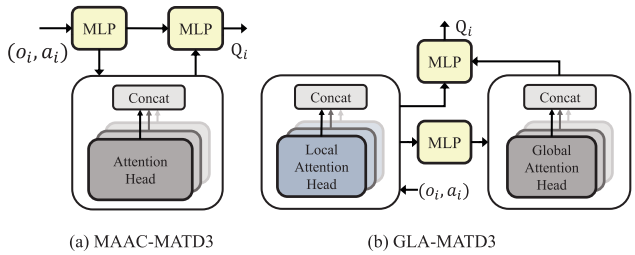
TABLE I
PERFORMANCE METRICS ACROSS DIFFERENT METHODS

Method	Scenario	Reward	Targets	Col-UAVs	Col-Obs
MATD3	4U-6O-12T	135.76 ± 23.96	16.79 ± 2.80	0.16 ± 0.05	0.62 ± 0.04
	6U-6O-15T	32.02 ± 29.51	7.13 ± 3.01	1.31 ± 0.10	0.95 ± 0.08
	8U-6O-18T	40.01 ± 16.37	6.88 ± 3.61	2.82 ± 0.14	1.69 ± 0.12
MAAC-MATD3	4U-6O-12T	139.05 ± 19.37	17.23 ± 1.61	0.17 ± 0.03	0.59 ± 0.03
	6U-6O-15T	109.06 ± 23.86	13.56 ± 2.64	0.58 ± 0.06	$\mathbf{0.63 \pm 0.08}$
	8U-6O-18T	79.54 ± 23.20	10.86 ± 2.97	1.78 ± 0.18	$\mathbf{1.15 \pm 0.17}$
GLA-MATD3	4U-6O-12T	139.95 ± 25.96	18.06 ± 2.36	0.14 ± 0.03	$\mathbf{0.42 \pm 0.03}$
	6U-6O-15T	138.10 ± 18.39	16.83 ± 1.72	0.43 ± 0.04	0.80 ± 0.09
	8U-6O-18T	94.26 ± 18.68	13.08 ± 2.84	0.86 ± 0.09	1.35 ± 0.14
SDA-MATD3	4U-6O-12T	$\mathbf{189.35 \pm 20.30}$	$\mathbf{21.78 \pm 1.69}$	$\mathbf{0.13 \pm 0.02}$	0.55 ± 0.05
	6U-6O-15T	$\mathbf{211.90 \pm 28.31}$	$\mathbf{24.95 \pm 3.07}$	$\mathbf{0.28 \pm 0.04}$	0.83 ± 0.06
	8U-6O-18T	$\mathbf{195.73 \pm 25.01}$	$\mathbf{22.62 \pm 2.58}$	$\mathbf{0.79 \pm 0.08}$	1.46 ± 0.16



(a) MAAC-MATD3 (b) GLA-MATD3

Fig. 4. Network architectures of the MAAC-MATD3 and the GLA-MATD3. The MAAC-MATD3 adopts MHA to process observations across agents. The GLA-MATD3 further extends the MAAC-MATD3 by incorporating an internal attention module that extracts important features from local observation of each agent prior to inter-agent communication.

three baselines in terms of cumulative reward and target coverage. These performance advantages are particularly pronounced in more complex environments. This highlights the effectiveness of the EAM with dynamic sparsity adjustment, which enhances local information extraction and enables better adaptability to dynamic and complex scenarios.

To further assess the performance of these four methods during distributed execution, we conducted 500 rounds of testing for the previously mentioned scenarios. The evaluation metrics included: 1) Average reward per UAV per round; 2) Average number of targets covered per UAV per round; 3) Number of UAV-to-UAV collision damages per round; 4) Number of UAV-to-obstacle collision damages per round. Table I summarizes the mean and standard deviation of the performance metrics for these four methods across these three scenarios, denoted as 4U-6O-12 T, 6U-6O-15 T, and 8U-6O-18 T. In Table I, the numbers in bold font denote the optimal values.

From Table I, our method shows clear advantages in average rewards and target coverage. In the most complex scenario (8U-6O-18 T), it achieves 3.29, 2.08, and 1.73 times the average target coverage, and 4.89, 2.46, and 2.08 times the average rewards compared to the three baselines. In the simplest scenario (4U-6O-12 T), collision rates are similar across methods. However, in more complex scenarios (6U-6O-15 T and 8U-6O-18 T), our method attains the lowest UAV-to-UAV collision rates (0.28 and 0.79), demonstrating superior coordination. UAV-to-obstacle collisions are higher due to stochastic obstacle movement, but our performance is comparable to baselines. Notably, with the map size fixed, as the environment complexity increases,

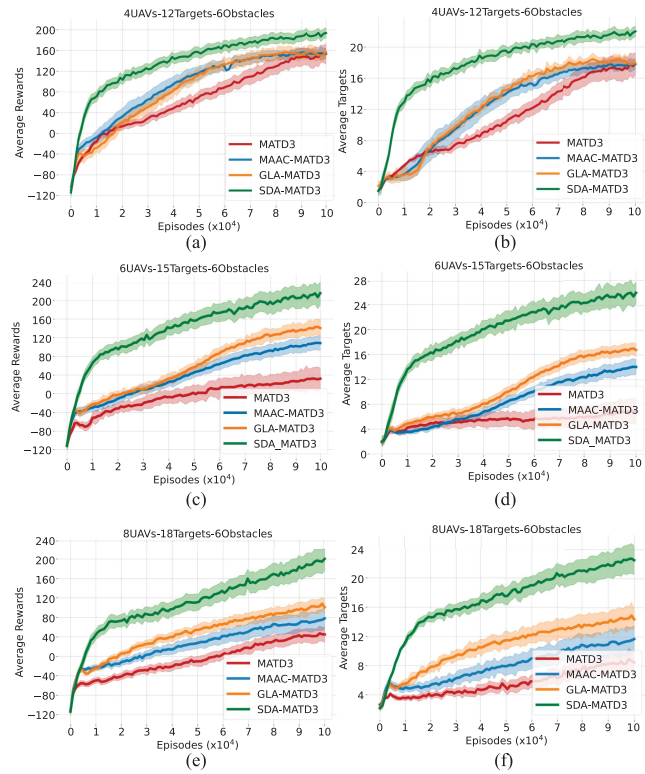


Fig. 5. Comparison of Average Reward Functions and Average Number of Covered Targets for Three Methods Across Three Scenarios (SDA-MATD3 is Our Method).

the metrics of the three baselines rapidly decline due to increased physical congestion and coordination difficulty, whereas our method shows only slight fluctuations, indicating strong robustness to increasing complexity.

C. Attention Analysis

To evaluate the ability of the EAM to learn attention distributions and the role of dynamic sparsity adjustment in filtering less relevant entities during training, we analyze the attention weights of the EAM at different training stages in the scenario with 8 UAVs, 18 targets, and 6 obstacles. Fig. 6 illustrates the attention weight distributions of each UAV toward other UAVs, obstacles, and targets after 30,000 and 100,000 training episodes.

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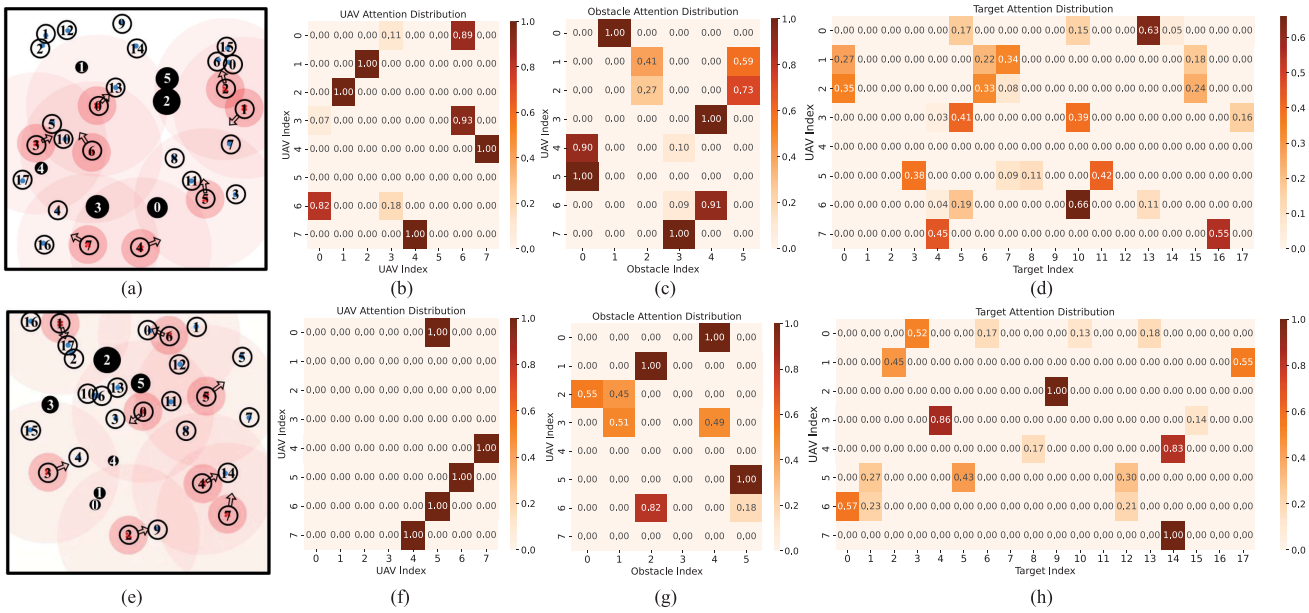

Fig. 6. Distribution of attention weights in the EAM after 30,000 and 100,000 training episodes. (a) and (e) depict UAVs in red, targets in blue, and obstacles in black. (b)-(d) and (f)-(h) illustrate the attention weights of UAVs assigned to different entity categories (the other UAVs, obstacles, and targets). The vertical axis represents the UAV index, while the horizontal axis denotes the index of the current entity category.

Fig. 6(a)–(d) and Fig. 6(e)–(h) illustrate the distribution of the attention weights for each UAV toward the other UAVs, obstacles, and targets after 30,000 and 100,000 training episodes, respectively. In general, UAVs tend to assign higher weights to closer entities with smaller angular deviations. After 30,000 training episodes, UAVs can correctly distinguish the relative importance of different entities, but the filtering of less relevant entities remains insufficient. After 100,000 episodes, when λ reaches its maximum value of 1, UAVs can not only differentiate the importance of entities effectively but also sparsify the attention weights of less relevant entities to zero.

Taking UAV U_0 as an example, after 30,000 training episodes, as shown in Fig. 6(d), although U_0 has learned to assign the maximum weight of 0.63 to target G_{13}, the attention to less relevant targets, such as G_{14}, has not yet been fully sparsified to zero. After 100,000 training episodes, in Fig. 6(e), U_0 detects targets G_3, G_6, G_8, G_{10}, G_{11}, G_{12}, and G_{13}. Considering its velocity direction and relative distances, the attention weights to G_8, G_{11}, and G_{12} are sparsified to zero, while G_3 receives the highest weight of 0.52. For obstacle attention, in Fig. 6(g), the weights for T_2 and T_5 are also sparsified to zero, while T_4, located in the velocity direction, receives the highest attention.

D. Ablation Studies

To assess the effectiveness of dynamic sparsity adjustment and the contribution of individual modules, we compare SDA-MATD3 with three ablated variants: SDA-MATD3 (w/o Sparse) using only Softmax, SDA-MATD3 (full Sparse) adopting Sparsemax entirely, and EAM-only-MATD3 without IAM. Fig. 7 presents the learning curves across three scenarios. We do not include an ablation removing EAM, as this variant is equivalent to the baseline MAAC-MATD3.

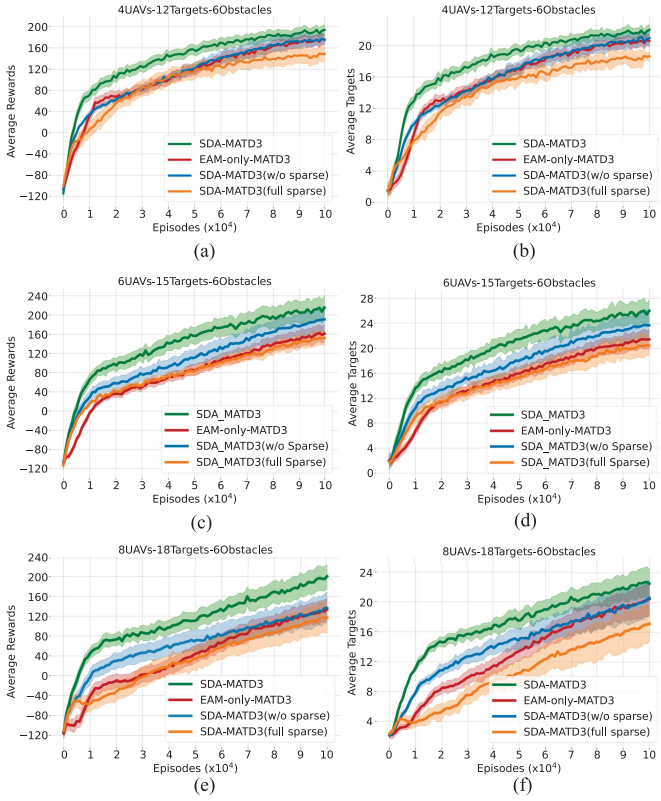

Fig. 7. Comparison of Average Reward and Average Number of Covered Targets Across Three Scenarios in Ablation Studies.

The comparison results demonstrate that, compared with the baseline using only the Softmax function or Sparsemax function, introducing the dynamic sparsity adjustment mechanism improves UAV coverage performance. In the early training

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stage, Softmax distributes attention weights evenly, preventing premature neglect of observed entities. As training progresses, UAVs learn the relative importance of different entities, and the sparsity control parameter gradually increases, allowing the EAM to focus on critical entities by sparsifying less relevant ones. The benefits of dynamic sparsity adjustment are especially pronounced in complex scenarios.In addition, the ablation study highlights the contribution of the IAM. Removing IAM (EAM-only-MATD3) weakens the inter-agent cooperation, leading to reduced coverage efficiency. The complete SDA-MATD3 consistently achieves the best performance, confirming the complementarity of IAM and EAM and their central role in enhancing cooperative UAV efficiency.

V. Conclusion

In this letter, we propose a dual-attention learning method with sparse optimization based on MADRL to address the cooperative coverage tasks in complex dynamic environments. Specifically, we introduce a hierarchical attention mechanism to handle different entity sets within local observations and incorporate a dynamic sparse adjustment mechanism. This allows agents to dynamically allocate attention weights to different entities and autonomously sparsify these weights, thereby enhancing the training efficiency and the coverage performance. The effectiveness and generalizability of the proposed method are validated through simulations in scenarios of varying complexity. Although the results are encouraging, the sparsity control parameter currently follows a simple linear schedule, which may not be optimal and could benefit from more adaptive or theoretically guided adjustment strategies. In future work, we will explore the practical deployment and application of MADRL in multi-UAV systems.

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